

Homework 5

Due Thursday, October 15 at 9:30 AM EDT

For this homework, turn in both a .ipynb file and a .pdf version.

Question One

This exercise is based on data that appeared in the November 9, 1988 edition of the *Wall Street Journal*. The example was originally used in Siegel (1997), but was also used in Sheather (2009). To quote Siegel:

US Treasury bonds are among the least risky investments, in terms of the likelihood of your receiving the promised payments. In addition to the primary market auctions by the Treasury, there is an active secondary market in which all outstanding issues can be traded. You would expect to see an increasing relationship between the coupon of the bond, which indicates the size of its periodic payment (twice a year), and the current selling price. The ... data set of coupons and bid prices (are) for US Treasury bonds maturing between 1994 and 1998 ... The bid prices are listed per “face value” of \$100 to be paid at maturity. Half of the coupon rate is paid every six months. For example, the first one listed pays \$3.50 (half of the 7% coupon rate) every six months until maturity, at which time it pays an additional \$100.

The data file can be found on Canvas.

Do each of the following. We will treat “Bid Price” as the response and “Coupon Rate” as the predictor.

1. Use the **seaborn** function **regplot** to create a scatter plot of the bid price versus the coupon rate with the least squares regression line superimposed. Comment on the form.
2. Fit a simple linear regression model to the data using Python.
3. Construct the plot of residuals versus fitted values and comment on its form.
4. Using the standard approach, construct a 95% confidence interval for β_1 based on this initial model. Do you trust this confidence interval?

Question Two

This exercise is based on the “Benchmark Bond Trade Price Challenge” which was conducted on the web site Kaggle.com. You can see the full details of this challenge at

<http://www.kaggle.com/c/benchmark-bond-trade-price-challenge>

but you may have to register (it is free). This is reproduced from the “Background:”

As far as price transparency is concerned, there has historically been a huge gap between the amount of reference information available to those trading equities versus those trading corporate bonds. Stock exchanges report trades, bids and offers at all times. Free access is available online with a 15 minute delay while traders who demand more

information can pay for ultra efficient real time data and information about size of current bids and offers. By contrast, bond trades are required to be reported within 15 minutes and only those who pay for the TRACE feed can access this information. No quotes are publicly available and the best way to get a quote is to solicit multiple brokers and wait for a reply. Alternatively there are data companies that provide end of day prices, published after the market has closed and with no guarantee that the specific information sought will be included. Accurate bond pricing is also hindered by lack of liquidity. Only a fraction of TRACE eligible bonds trade on a given day, so the most recent trade price is often multiple days old. Pricing bonds based on other more liquid bonds that have similar features is common, but again limited by the presence of such bonds.

Benchmark Solutions is the first provider of realtime corporate bond prices. Every 10 seconds we provide accurate prices that incorporate interest rate data, trades or quotes of the bond in question, trades or quotes of other bonds or CDS of the issuer of the bond in question as well as other input sources. Pricing bonds accurately requires an exacting knowledge of payment schedules, trading calendars and reference data for each bond. This, as well as synthesizing all of the bonds and CDS quotes and trades of a given issuer into implied hazard and funding curves, is something that we feel is beyond the scope of this challenge. Rather, we provide you with a reference price which is an intermediate result of our calculations and is labeled 'curve_based_price' in the dataset. Thus the competition focuses on trading dynamics and microstructure of individual bonds, rather than all bonds from a given issuer.

We created a reduced data set in order to make this challenge something we could manage during this exercise. This data set is available at

<http://www.stat.cmu.edu/~cschafer/MSCF/bonddata.txt>

We will fit regression models which attempt to model `trade_price`.

1. Read in the data. Make sure to correctly identify and transform any predictors which are categorical. Also, be sure to exclude any variables which are clearly not relevant as predictors in a regression, e.g., ID numbers.
2. Fit a model using all of the predictors. Construct the plot of residuals versus fitted values for this model, and comment on the quality of the fit.
3. Construct the normal probability plot of the residuals. What does this plot tell us?
4. Implement a stepwise variable selection procedure using AIC to find a simpler model. Report on the resulting model.
5. Repeat the above, but use PRESS instead of AIC. How does the resulting model change?